

Hybrid Photonic Integrated O-band PLC with SiN SSC and SOA Array Switches for Lossless Wavelength Selective Switching

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Abstract We demonstrate O-band lossless wavelength-selective switch using hybrid photonic-integrated U-shaped InP-SOA arrays and silica PLC AWGs with integrated SiN spot-size converters. Results show efficient coupling $<1.5\text{dB/facet}$, 12dB net-gain and WDM $100\text{G}/\lambda$ PAM4 switching with $>46\text{dB}$ on-off ratio and $<0.52\text{dB}$ power penalties at KP4.

Introduction

The continuous growth of data center traffic has driven the need for scalable, high-speed, energy-efficient, and configurable optical circuit switching (OCS) solutions to support increasing data rates while reducing the power consumption of optical/electrical/optical (O/E/O) conversions [1]. In this context, wavelength-division multiplexing (WDM) based switching enables high-capacity and flexible routing optical interconnects. In particular, wavelength selective switches (WSS) provide dynamic routing of individual channels for efficient utilization of network resources [2].

Current WSSs operate mainly in the C-band using micro-electromechanical systems (MEMS) or liquid crystal on silicon (LCoS) technologies [3, 4]. However, their large footprint, high cost, strict alignment requirements and millisecond switching speeds make them less suitable for datacenter networks [2]. Photonic integrated circuit (PIC) WSSs could offer compact footprint and cost-effective mass production.

Although O-band PIC based space switches have been demonstrated [5–7], only limited WSS works have been reported. Ring-resonator-based O-band WSS [8] suffers from high insertion losses of up to 25 dB, limiting their scalability. A key challenge in scaling PIC-based WSS lies in the accumulated optical loss.

Recently an $8 \times 8 \times \lambda$ wavelength cross-connect switch (WCS) based on broadcast and select architecture (Fig. 1(a)) implemented with discrete components was demonstrated [9]. It consists of passive $1 \times N$ splitters/combiners and WSS blocks based on AWGs (as WDM demux and mux) and SOA array. In [9], it was verified that mature silica planar lightwave circuit (PLC) AWGs provide low loss and flat-top operation with $<45\text{dB}$ cross-talk while discrete InP SOA array used as switches and amplifiers to compensate the WSS losses.

Hybrid photonic integration is a promising solution to realize high performance PIC- WSS blocks combining platforms with efficient InP SOAs that offer on-chip optical gain and fast switching capability with high on-off ratio, and PLC-based low-

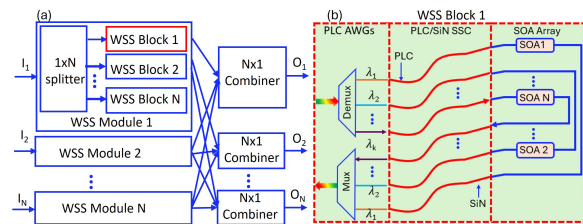


Fig. 1: (a) Modular $N \times N$ broadcast-and-select WSS architecture. (b) Schematic of wavelength-selective block.

loss, polarization-independent AWGs with excellent fiber coupling efficiency. However, efficient hybrid integration of these platforms remains challenging due to fundamental differences in waveguide properties. The significant mode-size mismatch between different platforms limits the efficiency of direct coupling.

In this work, we design, fabricate, and assess a hybrid integration platform that combines PLC/SiN spot-size converters (SSCs) with U-shaped SOA arrays based on InP passive-active integration. The SSC is realized by monolithically co-integration of silica PLC and SiN waveguides, acting as a photonic bridge between PLC and InP platforms (see Fig. 1(b)). It matches the large mode size of silica-based passive components (AWGs, splitters/combiners) while transitioning to a SiN waveguide optimized for coupling to the small mode size of InP SOAs. In [10], we demonstrated SSC–SOA coupling using straight waveguide interfaces. However, back-reflections induced undesired lasing of SOA, limiting the achievable SOA gain to a few dB, and resulting in no fiber-to-fiber gain after the SSC. In this work, we overcome this limitation by employing angled waveguide interfaces for both SSC and SOA, effectively suppressing back-reflections and enabling high-gain. As a result, we achieved a coupling loss below 1.5 dB per facet and an overall fiber-to-fiber net gain of up to 12 dB.

By integrating high-gain PLC/SSC–SOA array hybrid PICs with discrete PLC AWGs, we demonstrate net gain WSS block with WDM $100\text{G}/\lambda$ PAM4, achieving an on–off ratio exceeding 46 dB and a power penalty below 0.52 dB at KP4.

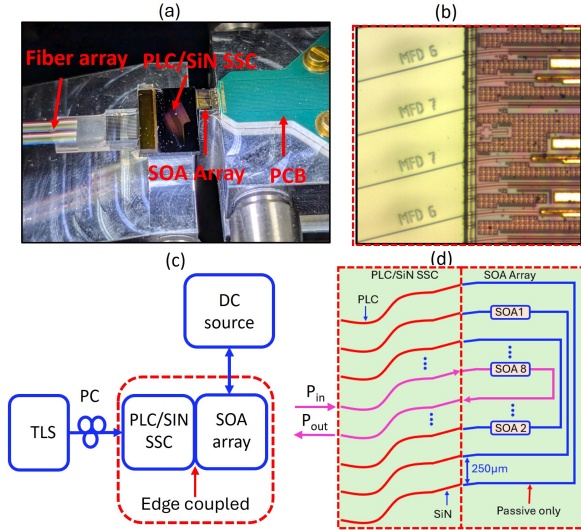


Fig. 2: (a) Fabricated InP SOA array and PLC/SiN SSC PICs. (b) Close-up of the PICs coupling interface. (c) Experimental setup to characterize the coupling and fiber-to-fiber gain. (d) Layout of the PLC/SiN SSC–SOA schematic coupling path.

Hybrid Integrated PLC/SiN SSC and SOA Array

The InP PIC SOA arrays are designed and realized using the active-passive photonic integration technology offered by Smart Photonics. The compact U-shaped geometry of the SOA array allows both input and output coupling on a single facet, thereby simplifying packaging and system complexity. The SOA array consists of 8 active SOAs resulting in 16 input/output waveguides spaced at 250 μm and an additional U-turn all-passive waveguide for calibration of the coupling loss. The waveguides feature a ridge structure with a 1.5 μm etch depth. The core has a width of 2 μm and a thickness of 0.4 μm , with an active section length of 900 μm . The waveguides are tilted at 7° near the facet to suppress back-reflections.

The SSC is fabricated through monolithic integration of silica PLC and SiN waveguides stacked vertically. On the PLC side, it consists of a pure silica waveguide with a core thickness and width of 4.2 μm , enabling efficient mode matching to standard SMFs and other glass waveguides. Toward the SOA side, a SiN waveguide layer is introduced beneath the silica, where the optical mode gradually transitions into the higher-index SiN region and becomes confined. The SiN waveguide, with a thickness of 0.2 μm and width of 2.5 μm , reduces the mode size for efficient coupling to InP waveguides. The SSC waveguides are spaced 250 μm and are fiber-pigtailed on the PLC side. The SiN waveguides are tilted by 15° to allow efficient coupling with the angled SOAs.

Fig. 2(a) shows the fabricated PICs mounted on separate holders. The SOA array is wire-bonded to a printed circuit board (PCB) for individual current bias control. Fig. 2(b) presents a close-up of PICs. To evaluate the coupling efficiency and

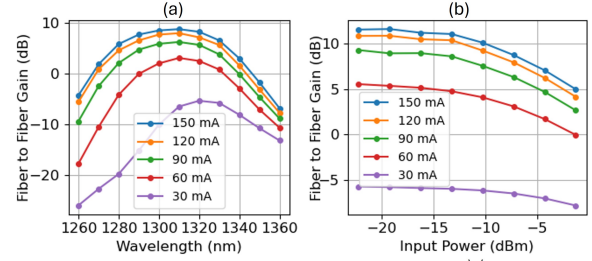


Fig. 3: (a) Fiber-to-fiber gain across the O-band. (b) Gain vs input power at 1310 nm.

fiber-to-fiber gain, we employ the setup shown in Fig. 2(c, d). The pigtailed SSC chip is placed on a custom vacuum chuck mounted on a 4-axis stage (y, z, θ_y, θ_z). The SOA array is glued on a water-cooled (20 °C) aluminium holder and placed on a 6-axis stage ($x, y, z, \theta_x, \theta_y, \theta_z$) as shown in Fig. 2(a). The input signal from a tunable laser source (TLS) is launched into a pigtailed SSC input fiber, coupled to the corresponding SOA, routed through a U-turn path, and returned to the SSC output fiber (magenta path in Fig. 2(d)). The input and output powers ($P_{\text{in}}, P_{\text{out}}$) are measured at the input and output SSC fibers to assess SOA gain and coupling loss.

Fig. 3(a) reports the measured fiber-to-fiber gain across the O-band for one SOA as a representative under different bias currents and wavelengths at $P_{\text{in}} = -7.2$ dBm. The smooth gain evolution with increasing current confirms suppression of back-reflection and lasing ripple due to the angled waveguide interfaces. Fig. 3(b) shows the gain characteristics at 1310 nm for different input powers. A net fiber-to-fiber gain of up to 12 dB is achieved at low input powers, with similar performance for the other SOAs. From the coupling loss of the passive U-turn paths, coupling loss is estimated to be below 1.5 dB per facet.

WSS Block Implementation and Assessment

With the demonstrated high gain and efficient coupling, the SSC–SOA platform is further integrated with a demux and mux AWGs to realize the WSS block with net gain.

The proposed modular WSS architecture employs a broadcast-and-select configuration, as illustrated in Fig. 1(a). In this scheme, $1 \times N$ splitters distribute the input signal into N parallel WSS blocks, each associated with a different output port by $N \times 1$ combiners. This enables any wavelength channel from any input ($I_1, I_2, \dots, I_n$) to be routed to any output ($O_1, O_2, \dots, O_n$), providing full connectivity and flexible resource allocation. Since all modules are identical, the overall system performance can be inferred from a single WSS block.

As shown in Fig. 1(b), each WSS block consists of a silica PLC-based AWG demultiplexer, a hybrid PLC/SiN SSC and InP SOA array, and a second

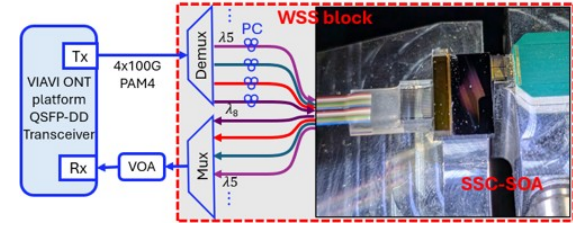


Fig. 4: Data transmission measurement setup. PC: polarization controller.

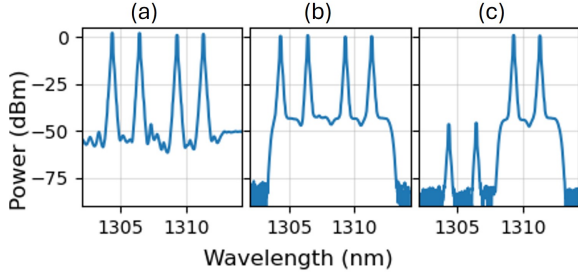


Fig. 5: (a) Spectrum after VIAVI TX. (b) Spectrum at the output of WSS block when all channels are selected. (c) Selective switching of two channels showing >46 dB on-off ratio

AWG as multiplexer. The AWGs are designed with a polarization-independent flat-top spectral response to reduce sensitivity to laser wavelength drift and minimize inter-channel crosstalk. They support 8 channels spanning 1295.5–1311.6 nm with a uniform 400 GHz spacing (2.3 nm) and a 1-dB bandwidth of 1.7 nm. The measured insertion loss is approximately 3.5 dB with inter-channel crosstalk below -45 dB, making them suitable for low-loss and scalable WSS operation [9].

As shown in Fig. 3(a), the gain profile of the SSC-SOA is quite flat across the 8 channel of the AWG (1295–1320nm), ensuring uniform amplification of all channels. The achieved net gain not only compensates for the mux and demux AWGs loss (7dB in total) but can also be used to partially compensate for the splitter/ combiner loss.

To validate the performance of WSS block, we conducted data transmission and switching experiment using the setup shown in Fig. 4. The transmitter consists of a commercial 400G QSFP-DD transceiver hosted on a VIAVI ONT platform, which generates four O-band lanes at 1304.35 nm, 1306.43 nm, 1309.35 nm and 1311.35 nm. These lanes, which align with the last four AWG channels ($\lambda_5 - \lambda_8$), utilize PAM4 modulation for an aggregate data rate of 400 Gb/s (100G/lane). The measured launch powers at the transceiver output are 3.3, 3.5, 2.4, and 2.7dBm, respectively.

As the available transceiver supports only four channels, we measure only four of the eight available channels. The four TX lanes are demultiplexed by the first AWG, traverse the SSC-SOA U-path, and are subsequently recombined by the second AWG. Four SOAs are biased simultaneously via a multi-channel DC source.

The spectral evolution of the signal at various stages is shown in Fig. 5(a–c). Fig. 5(a) shows

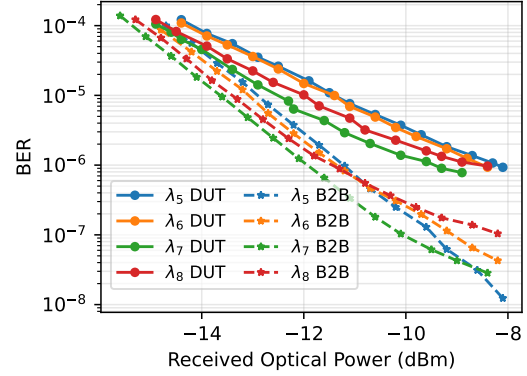


Fig. 6: Measured BER for the four PAM4 lanes vs. B2B.

the initial spectra of the 4 channels after the transmitter, while Fig. 5(b) displays the spectra at the WSS block output when all SOAs are in the “on” state. It can be seen that the gain of the SOA array restores the signal power, demonstrating the loss-less WSS block. Another switching configuration to measure the on-off ratio of the SOA array is illustrated in Fig. 5(c). By selectively biasing SOAs at 0 mA (“off” state), the first two channels are blocked while the last two remain in the “on” state. The blocked channels are below SOA ASE noise with an ER > 46 dB. This is essential to avoid wavelength contention in $N \times N$ WCS, when a specific wavelength channel of only one selected input port should be routed to a common output, the SOA gates in the other WSS blocks must effectively block that channel to prevent signal interference.

For BER measurement, we used variable optical attenuator (VOA) to vary the received optical power. The performance of BER was evaluated simultaneously for all four channels compared to a back-to-back (B2B), as shown in Fig. 6. The measured BERs for all channels are below the standard KP4-FEC threshold of 2.4×10^{-4} [11]. The power penalties for λ_5 through λ_8 are 0.51, 0.5, 0.52, and 0.44dB, respectively, at BER 1×10^{-4} . These results confirm that the photonic integrated WSS block provides stable, good signal quality, demonstrating its robustness for scalable, high-capacity O-band optical networks.

Conclusions

We demonstrate a hybrid integration of PLC/SiN SSCs with InP SOA arrays, achieving < 1.5 dB coupling loss per facet and up to 12 dB fiber-to-fiber gain, sufficient to compensate passive losses in modular WSS block architectures. Using this platform, we realize an O-band WSS block and validate it with 400G PAM4 signals, achieving > 46 dB extinction ratio and < 0.52 dB power penalty. These results show the potential of this approach for scalable, low-loss photonic integrated WSS in data center networks.

Acknowledgments

The work is partially supported by the Dutch National Growth Fund for Photonics ‘PhotonDelta’ and by the EU research and innovation program ADOPTION grant agreement 101070178. The authors also thank Dr. Robert Palmer and Dr. Giuseppe Talli from Huawei for providing the AWGs.

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